WORKED SOLUTION

STEP II 1988, Question 3

Pure • Symmetries of Roots

STEP II 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q3

The key idea

Don't work with the coefficients first; work with the set of roots. The property says the set of roots is unchanged by the operation $k \mapsto 1/k$, so either the two roots swap or each is fixed. Adding the second operation $k \mapsto 1 - k$ does the same job again. For the cubic, applying the two operations over and over produces up to six values from one root, and a cubic has room for only three: the roots must sit at special points where some of the six values coincide.

Solution

The reciprocal property

Let the roots be α and β (possibly equal, possibly complex), so $\alpha + \beta = -b$ and $\alpha\beta = c$. First, 0 can't be a root, because 0^{-1} doesn't exist; so $c \neq 0$.

The map $k \mapsto 1/k$ must send each root to a root, so $1/\alpha \in \{\alpha, \beta\}$.

- If $1/\alpha = \beta$, then $\alpha\beta = 1$, that is, $c = 1$. Conversely, if $c = 1$ the roots are nonzero with $\beta = 1/\alpha$, so the reciprocal of each root is the other root, and the property holds for *every* real b. (This includes repeated roots ± 1 when $b = \mp 2$, and complex roots on the unit circle when $|b| < 2$.)
- If $1/\alpha = \alpha$ but $1/\alpha \neq \beta$, then $\alpha = \pm 1$ and $\alpha\beta \neq 1$. Applying the property to β, its reciprocal is not α (since $\alpha\beta \neq 1$), so $1/\beta = \beta$ and $\beta = \pm 1$ too. With $\alpha\beta \neq 1$ the roots must be 1 and -1, giving $x^2 - 1 = 0$: $b = 0$, $c = -1$. This equation does have the property.

So the property holds exactly when

$$c = 1 \ \ (b \text{ any real number}), \qquad \text{or} \qquad b = 0,\ c = -1.$$

Adding the property $k \mapsto 1-k$

The same argument with $1-k$ in place of $1/k$: either $1-\alpha=\beta$, so $\alpha+\beta=1$; or both roots are fixed, $1-\alpha=\alpha$ and $1-\beta=\beta$, so $\alpha=\beta=\frac{1}{2}$ and again $\alpha+\beta=1$. Either way $b=-1$, and conversely $b=-1$ makes the roots α and $1-\alpha$, so the property holds.

The case $b=0$, $c=-1$ fails this, so we need $c=1$ and $b=-1$:

$$b=-1, \qquad c=1,$$

giving $x^2-x+1=0$ with roots $\frac{1}{2}\pm\frac{\sqrt{3}}{2}\mathrm{i}$. Each has modulus 1 and real part $\frac{1}{2}$, so $1/k$ and $1-k$ are both the complex conjugate $\bar{k}$, the other root.

The cubic

Now let $x^3+px^2+qx+r=0$ have real coefficients and both properties. As before 0 is not a root, and nor is 1 (else $1-1=0$ would be one).

A cubic with real coefficients has at least one real root; call it a. Applying the two operations repeatedly, each of the following is a root:

$$a, \quad \frac{1}{a}, \quad 1-a, \quad \frac{1}{1-a}, \quad 1-\frac{1}{a}=\frac{a-1}{a}, \quad \frac{1}{(a-1)/a}=\frac{a}{a-1}.$$

(The fourth is the reciprocal of the third, the fifth is $1-$ the second, the sixth the reciprocal of the fifth.) These are six real numbers, but the cubic has at most three distinct roots, so two of them coincide.

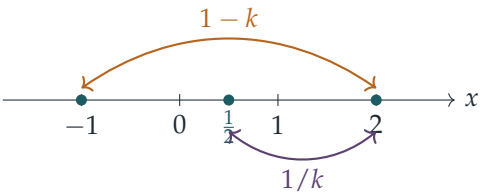
? **Ask yourself** How does "two of these are equal" pin down a?

Call the six maps $f_1,\dots,f_6$, with f_1 the identity. Each one has an inverse in the list: $\frac{1}{k}$, $1-k$ and $\frac{k}{k-1}$ undo themselves, while $\frac{1}{1-k}$ and $\frac{k-1}{k}$ undo each other. One can also check that composing any two of the maps gives another in the list. So if $f_i(a)=f_j(a)$ with $i\neq j$, applying f_i^{-1} gives $a=g(a)$, where $g=f_i^{-1}\circ f_j$ is one of the six and is not the identity. So a is a fixed point of one of the five non-identity maps:

$$\begin{aligned} \tfrac{1}{a}=a &\iff a=\pm1, & \tfrac{1}{1-a}=a &\iff a^2-a+1=0,\\ 1-a=a &\iff a=\tfrac{1}{2}, & \tfrac{a-1}{a}=a &\iff a^2-a+1=0,\\ \tfrac{a}{a-1}=a &\iff a(a-2)=0. \end{aligned}$$

The quadratic a^2-a+1 has no real roots (it is the quadratic from before), and $a\neq0,1$. Hence $a\in\{-1,\frac{1}{2},2\}$.

Whichever of these a is, the six values above run through exactly -1, 2 and $\frac{1}{2}$: the map $1-k$ swaps -1 and 2 and fixes $\frac{1}{2}$, while $1/k$ swaps 2 and $\frac{1}{2}$ and fixes -1.



So -1, 2 and $\frac{1}{2}$ are three distinct roots of the cubic. A monic cubic is determined by its three roots, so it must be

$$(x+1)(x-2)\left(x-\tfrac{1}{2}\right) = (x^2-x-2)\left(x-\tfrac{1}{2}\right) = x^3 - \tfrac{3}{2}x^2 - \tfrac{3}{2}x + 1.$$

Conversely, this cubic does have both properties, since its root set $\{-1, \frac{1}{2}, 2\}$ is sent to itself by $k \mapsto 1/k$ and by $k \mapsto 1-k$. So

$$x^3 - \tfrac{3}{2}x^2 - \tfrac{3}{2}x + 1 = 0$$

is the only such cubic, as required.

Checks

Substitute $x = 2$: $8 - 6 - 3 + 1 = 0$; $x = -1$: $-1 - \frac{3}{2} + \frac{3}{2} + 1 = 0$. The product of the roots is $-1 \cdot 2 \cdot \frac{1}{2} = -1 = -r$, and their sum is $\frac{3}{2} = -p$. Notice the cubic's coefficients read the same forwards and backwards up to sign $(1, -\frac{3}{2}, -\frac{3}{2}, 1)$: that palindromic shape is exactly what $k \mapsto 1/k$ forces. A computer test of 20,000 random (b, c) confirmed the reciprocal condition, and random real a away from $0, \pm 1, \frac{1}{2}, 2$ always gave six distinct values.

Common slips

- Answering only "$c = 1$" in the first part: the case where each root is its own reciprocal adds $b = 0$, $c = -1$.
- Overlooking the pair $\frac{1}{2} \pm \frac{\sqrt{3}}{2}\mathrm{i}$, which is closed under both maps. Starting from a *real* root is what rules it out: a real cubic has a real root, and that root's six values can't include these.
- Proving only that the given cubic works; the question asks you to show it is the *only* one.

Beyond STEP

The six maps $k \mapsto k$, $1/k$, $1-k$, $\frac{1}{1-k}$, $\frac{k-1}{k}$, $\frac{k}{k-1}$ form a group under composition, a copy of the symmetric group S_3; it permutes the three points $0, 1, \infty$, and the six values are the possible cross-ratios of four points. The counting step is the orbit–stabiliser

theorem: an orbit is shorter than six only at points fixed by some non-identity element. Allenby's *Rings, Fields and Groups* develops groups acting on sets, and Needham's *Visual Complex Analysis* shows these maps as Möbius transformations of the plane.

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